Why the Model Predicts a Complaint Category

1. Aim and model examined

This analysis explains the saved Linear SVM (C=1), the model selected using validation macro-F1. It examines learned word and phrase weights across all nine categories, then computes a local explanation for one new illustrative complaint. The local example is synthetic, not a held-out test record or an additional accuracy evaluation.

2. Explanation method

The TF-IDF vectorizer converts cleaned text into weighted word and two-word features. For each category c, the linear classifier calculates:
score(c) = intercept(c) + sum of [TF-IDF(feature) x weight(c, feature)]
The category with the highest score is predicted. A positive contribution raises that category's score; a negative contribution lowers it. Absent features contribute zero. The intercept also affects the score.

Global coefficients describe learned associations; a large weight alone does not imply a large contribution in every complaint. Local contributions depend on the feature's TF-IDF value in the specific input. This is direct linear-model inspection, not SHAP and not a causal explanation.

3. Strongest positive features by category

Category	Top three features (coefficient)
Bank accounts	checking (1.882); bank (1.866); chime (1.724)
Credit cards	card (3.043); credit card (2.272); interest (1.527)
Credit reporting	these (1.586); equifax (1.493); report (1.468)
Debt collection	debt (3.130); collection (1.916); owe (1.354)
Money services	paypal (1.506); cash (1.453); cash app (1.328)
Mortgages	mortgage (3.392); escrow (1.817); my mortgage (1.535)
Payday/title/personal loans	loan (1.179); affirm (1.070); the loan (1.057)
Student loans	mohela (1.915); student (1.664); loans (1.405)
Vehicle loans	car (1.939); vehicle (1.878); permission to (1.204)

Interpretation

Product-specific terms make many decisions understandable. However, learned vocabulary may also encode company-specific or sample-specific associations. For example, "chime" is among the strongest bank-account features. Plausible words do not establish that the model is reliable for every category.

A Measured Local Explanation

Illustrative input: My mortgage servicer applied my payment incorrectly and charged a late fee.

Predicted category: Mortgages

Category score summary	Value
Predicted-class decision score	0.627661
Intercept	-1.010147
Sum of all active feature contributions	1.637808
Runner-up category	Credit cards

4. Largest contributions by absolute magnitude

Feature	TF-IDF	Class weight	Contribution
mortgage	0.2497	3.3917	+0.8470
my mortgage	0.2976	1.5347	+0.4567
servicer	0.3067	0.4229	+0.1297
fee	0.2584	0.4433	+0.1145
applied	0.2248	0.3509	+0.0789
payment	0.1407	0.3162	+0.0445
my payment	0.2411	0.1546	+0.0373
charged late	0.3923	-0.0780	-0.0306

The table shows the eight largest absolute contributions; remaining active features sum to -0.040218. Including all features and the intercept reproduces the saved model's decision score (checked numerically). Individual displayed values are rounded.

5. What the explanation establishes

The contributing terms identify which recognized words and phrases raised or lowered the mortgage score. This supports transparent inspection of this prediction. A contribution is measured in decision-score units, not percentage confidence. Contributions shown are for the predicted category; they do not fully explain its margin over another category.

6. Limitations and conclusion

Unigrams and overlapping bigrams can share evidence. TF-IDF normalization makes contributions input-dependent. Negation, sarcasm, spelling variations and unfamiliar language may be poorly represented. Explanations reflect the model's learned associations, not proof of a consumer's intended meaning. The analysis provides faithful additive score explanations, but must be considered alongside the separate performance audit.

Reproducibility and evidence: Load outputs/model.joblib, clean the displayed input with src/project.py:clean, transform with the saved tfidf step, and multiply active feature values by the predicted-class coefficient row. Global values are from outputs/top_features.csv. No retraining was performed for this report.